

Preface

Civilization has passed the threshold of the second industrial revolution. The first industrial revolution, which was based upon the steam engine, enabled man to multiply his physical capability to do work. The second industrial revolution, which is based upon semiconductor electronics, is enabling man to multiply his intellectual capabilities. VLSI (Very Large Scale Integration) electronics, the most advanced state of semiconductor electronics, represents a remarkable application of scientific knowledge to the requirements of technology. This treatise is published in recognition of the need for a comprehensive exposition that describes the state of this science and technology and that assesses trends for the future of VLSI electronics and the scientific base that supports its development.

These volumes are addressed to scientists and engineers who wish to become familiar with this rapidly developing field, basic researchers interested in the physics and chemistry of materials and processes, device designers concerned with the fundamental character of and limitations to device performance, systems architects who will be charged with tying VLSI circuits together, and engineers concerned with utilization of VLSI circuits in specific areas of application.

This treatise includes subjects that range from microscopic aspects of materials behavior and device performance—through the technologies that are incorporated in the fabrication of VLSI circuits—to the comprehension of VLSI in systems applications.

The volumes are organized as a coherent series of stand-alone chapters, each prepared by a recognized authority. The chapters are written so that specific topics of interest can be read and digested without regard to chapters that appear elsewhere in the sequence.

There is a general concern that the base of science that underlies integrated circuit technology has been depleted to a considerable extent and is in need of revitalization; this issue is addressed in the National Research Council (National Academy of Sciences/National Academy of

Engineering) report entitled “Microstructure Science, Engineering and Technology.” It is hoped that this treatise will provide background and stimulus for further work on the physics and chemistry of structures that have dimensions that lie in the submicrometer domain and the use of these structures in serving the needs of humankind.

I wish to acknowledge the able assistance provided by my secretary, Mrs. Lola Goldberg, throughout this project, and the contribution of Academic Press in preparing the index.